

Gurpreetsingh Saini

8553 Manor Avenue, Apt C
Munster, IN 46321

E-mail:gsaini.uic@gmail.com
Phone: 312-623-1543

OBJECTIVE

Seeking a challenging position in an organization that would utilize as well as enhance my technical and management skills

PROFESSIONAL SUMMARY

- Currently working as a Quality Manager handling a team of six members.
- ASQ Certified Six Sigma Green Belt
- ASQ Certified Quality Engineer
- ISO 9001:2008 Internal Quality System Auditor: RABQSA QM & AU module competent
- 1.5 years of experience in the field of Machine Analysis: Caterpillar Excavator Hydraulics and Control systems
- Extensive experience with software like Matlab-Simulink, Dynasty, DATK and Pro/Engineer
- 1+ years of experience in the field of Computer Aided Design

ACADEMICS

University of Illinois at Chicago (UIC) Master of Science in Mechanical Engineering (May 2010)	GPA: 3.88/4.0
Gujarat University (G.U), Ahmedabad, Gujarat, India Bachelor of Engineering (B.E.) in Mechanical Engineering (July 2007)	GPA: 3.98/4.0

SKILL SET

Caterpillar Inc. Tools: Dynasty, Data Analysis Tool Kit (DATK)

Programming Language: Matlab – Simulink, C

CAD Tools: Pro/Engineer, SolidWorks, AutoCAD

FEA Tools: Ansys Workbench

Statistical Tools: MS Excel, Minitab

TECHNICAL WORK EXPERIENCE

Quality Manager (Dec '11-present)
B&B Manufacturing Inc.,

- Apply principles of lean manufacturing like value stream mapping, developing leader standard work, visual controls, etc in order to eliminate waste and maintain flow through the shop floor.
- Conduct Advance Product Quality Planning (APQP), write control plans and co-ordinate with cross functional team in order to obtain all the documents for Production Part Approval Process (PPAP)
- Handle customer complaints: conduct root cause analysis, implement Corrective/Preventive Actions (CAPA) and sustain those actions. Primarily use Define-Measure-Analyze-Improve-Control (DMAIC) approach to eliminate any systemic errors in the process that cause Non-Conformances
- Work with domestic and overseas suppliers to resolve any quality issues. Give objective inputs to the purchasing team in order to seek Return Material Authorizations when needed.
- Maintain and report on various metrics in a Plan-Do-Check-Act format focusing on Safety, Quality, Delivery, Cost and Cash Flow not only within QC department level but also on complete organization level.
- Work as a Lead Auditor as well as Document Control Coordinator for ISO 9000 activities

Mechatronics Engineer/Machine Analyst (May '10-Dec '11)
Servo Tech Inc. / Caterpillar Inc.

- Integrated various Engine and Pump models in complete HEX Machine Model (Dynasty) and updated related controls (Simulink) on daily basis.

- Conducted L5 level tests like P-Q, Boom-Up Power checks, 2-pump stall and L6 level tests like 90° Truck Loading and resolved any controls or Machine Model issues. Held clear communications with Engine and Pump team in order to obtain objective inputs on debugging related issues.
- Write Matlab codes to automate tedious analysis activities like processing test data to feed into Dynasty Ascii or Data Interpolator components, performance analysis, etc.

Independent Pump Controls:

- Explored the opportunity of and developing a control strategy for running drive and idler pumps based on two independent torque based signals to achieve better cycle efficiency. This project involved thorough understanding of Excavator Hydraulics, its controls and the analysis of 90° truck loading cycle in order to design the control strategy using Simulink.

Validate the Simulation Model to the 336X T4F Testbed Excavator Machine:

- Calibrated the Soil Model to Test Data and developed Operator Model for Same Level 90° Truck Loading for North American Linkage Configuration (longer stick and bigger bucket). Extensive use of Dynasty and DATK software and was involved during this project.
- This validated model was used as a Baseline model for new concept evaluation.

Teaching Assistant

(Jan '10 – Apr '10)

Computer Aided Design (CAD), Mechanical and Industrial Engineering Department, UIC

- Teach senior and graduate students Pro/engineer.
- Help students understand the topics like transformations and manipulation of objects, description of curves, finite element methods, etc.
- Set and correct homework and test problems.

Research Assistant

(Jan '09 – Apr '10)

Biomechanics Lab, University of Illinois at Chicago (UIC)

- Studied the effect of stem length and porous coating length of the Hip Prosthesis on the stresses induced in the femur bone.
- Generated parametric models of the Hip Prosthesis and the femur bone on Pro/Engineer
- Carried out Finite Element Analysis on these models under the loading conditions of body weight and abductor muscles using Ansys workbench.

Graduate Engineer Trainee

Crompton Greaves Ltd, Light Source Division, Baroda, Gujarat, India

(Jul '07 - Nov '07)

- Worked in production department of Compact Fluorescent Lamps (CFL) – was a member of a team looking after phosphorous coating of the lamps.
- Designed a tray that could be installed on the table of coating machine so that the spilled and surplus coating suspension from the freshly coated tubes would directly flow into the tank so that it can be reused.

Quality Engineer

(Dec '06 – July '07)

Triveni Enterprise, Baroda, Gujarat, India

- Perform daily inspection activities including first article inspections, in process inspection checks and final inspections.
- Point of contact for customers and suppliers for quality issues
- Introduced the company to Six Sigma problem solving techniques, interacted with team of CNC programmers and technicians to apply DMAIC methodology to result in 40% saving on the machining cost of a component.

ACADEMIC PROJECTS AND PRESENTATIONS

- Presented a poster on “**Effect of stem length and extent of porous coating on stress shielding in total hip arthroplasty**” at Orthopedic Research Society 56th Annual Meeting at New Orleans, Louisiana **(March '10)**
- **Position and speed control** of a **DC Motor** using PIC Microcontroller (18F4331) and a self designed encoder.
- **Position control of an inverted pendulum** on a moving body, using Matlab.

Keywords: Masters, Mechanical, Dynasty, DATK, Matlab, Simulink, ProEngineer, SolidWorks, Ansys, Finite Element Analysis, Six Sigma, CAD, CAM